

to get proper oscillation frequency.

In this circuit, the devices are controlled through its port pins P1.0 through P1.7. The controlling is done by the user from the PC with the

Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	High when S1 is pressed
TP3	Low when device1 is on

user interface shown in Fig. 7. The microcontroller is connected to the PC through MAX232 driver as shown in Fig. 2.

MAX232. The serial output of the computer is connected to MAX232 driver IC2 through DB9 connector as shown in Fig. 2. The MAX232 IC is used for the long-distance and error-free transmission of data. It has two internal charge-pumps that convert +5V into $\pm 10V$ for RS232 driver operation. Pins 11 and 12 of IC2 are

connected to pins 3 and 2 of microcontroller IC1, respectively. Serial data command from the PC is received by microcontroller IC1 through driver IC2. The microcontroller then triggers independent opto-TRIAC drivers for devices corresponding to the command received.

An RS232 serial port was once a standard feature of a personal computer, used for connections to modems, printers, mice, data storage, uninterruptible power sup-

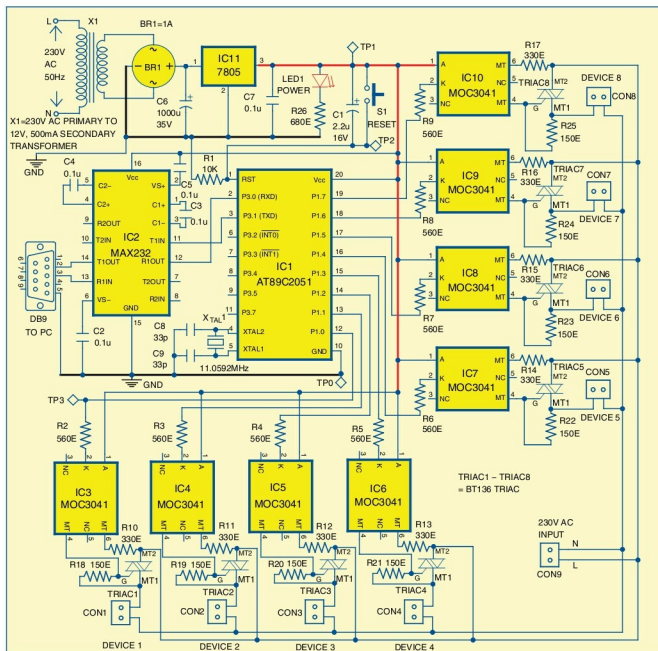


Fig. 2: Circuit of PC-based equipment controller